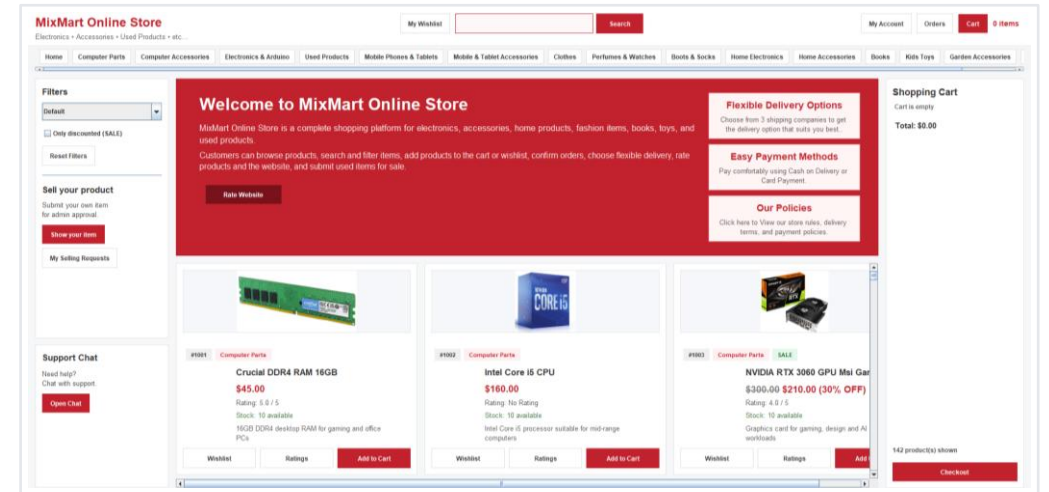


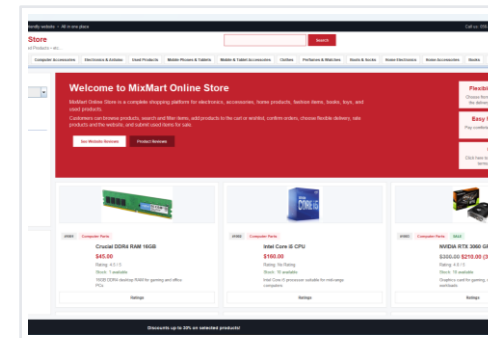
# Mix Mart Online

## Online Shopping Store Database Project

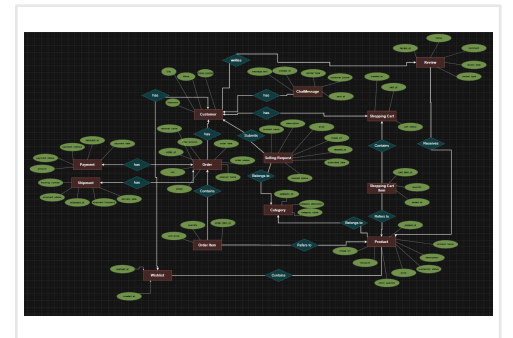
Java Swing + MySQL + JDBC



Customer home screen



Admin home



ERD

Prepared by: Melad Sam (12323283) and Emad Ataout (12322222)  
Course: Database Systems | An-Najah National University | Spring 2026

# Project Idea & Problem

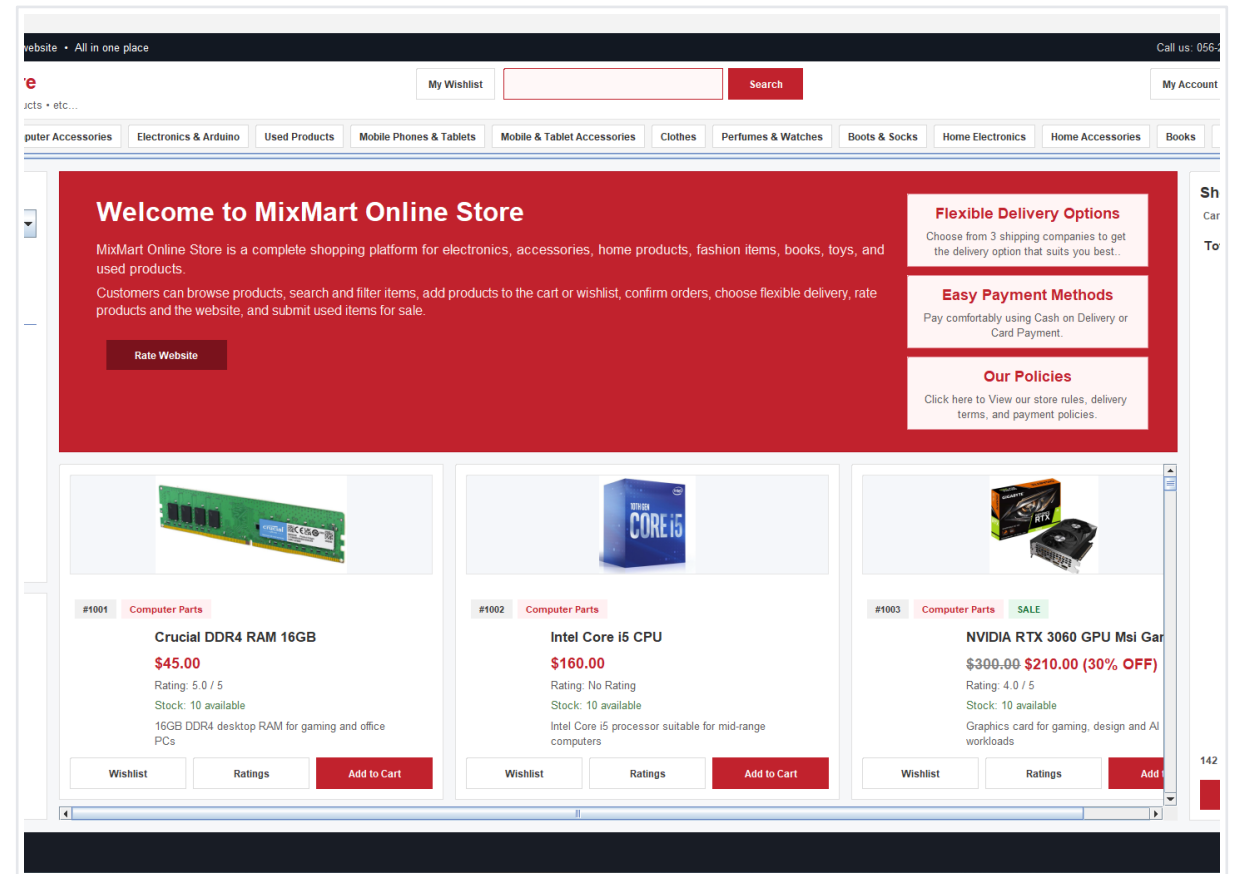
Why this database-backed store was built

Online stores need organized product, order, stock, payment, and customer data.  
Manual or file-based storage can cause repeated data, missing orders, and stock mistakes.  
Mix Mart Online connects a real Java Swing GUI with a MySQL database to support daily store operations.

## Project result

A working desktop shopping system with customer workflows, admin workflows, database relationships, validations, and testing screenshots.

Mix Mart Online | Final Project Demo



Home page: products, filters, cart, categories

# System Architecture & Technologies

Simple layered structure: GUI → DAO/JDBC → MySQL

## Frontend

Java Swing desktop interface for forms, buttons, dialogs, product cards, cart sidebar, and admin panels.

## Connection Layer

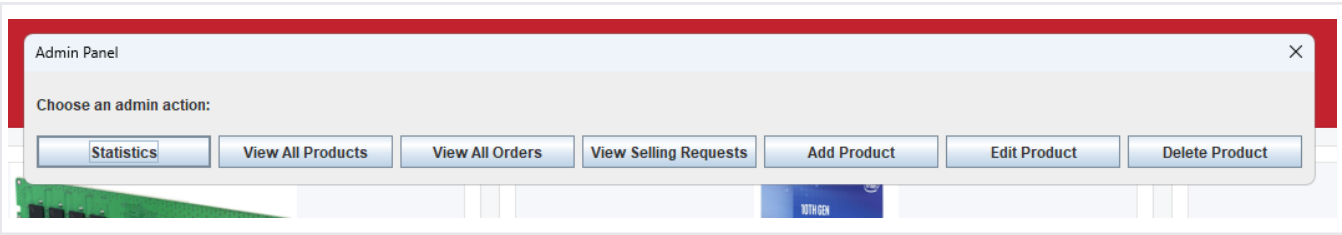
DAO classes separate SQL operations from GUI code. JDBC connects Java with MySQL and executes queries.

## Database

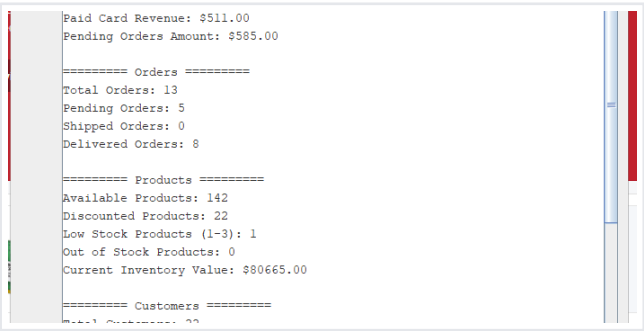
MySQL database: online\_shopping\_store\_database. Stores customers, products, carts, orders, payments, shipments, reviews, selling requests, and chat.

## Key DAO classes

CustomerDAO  
ProductDAO  
CartDAO  
OrderDAO  
ReviewDAO  
WishlistDAO  
SellingRequestDAO  
ChatDAO  
AdminStatsDAO



Admin panel navigation



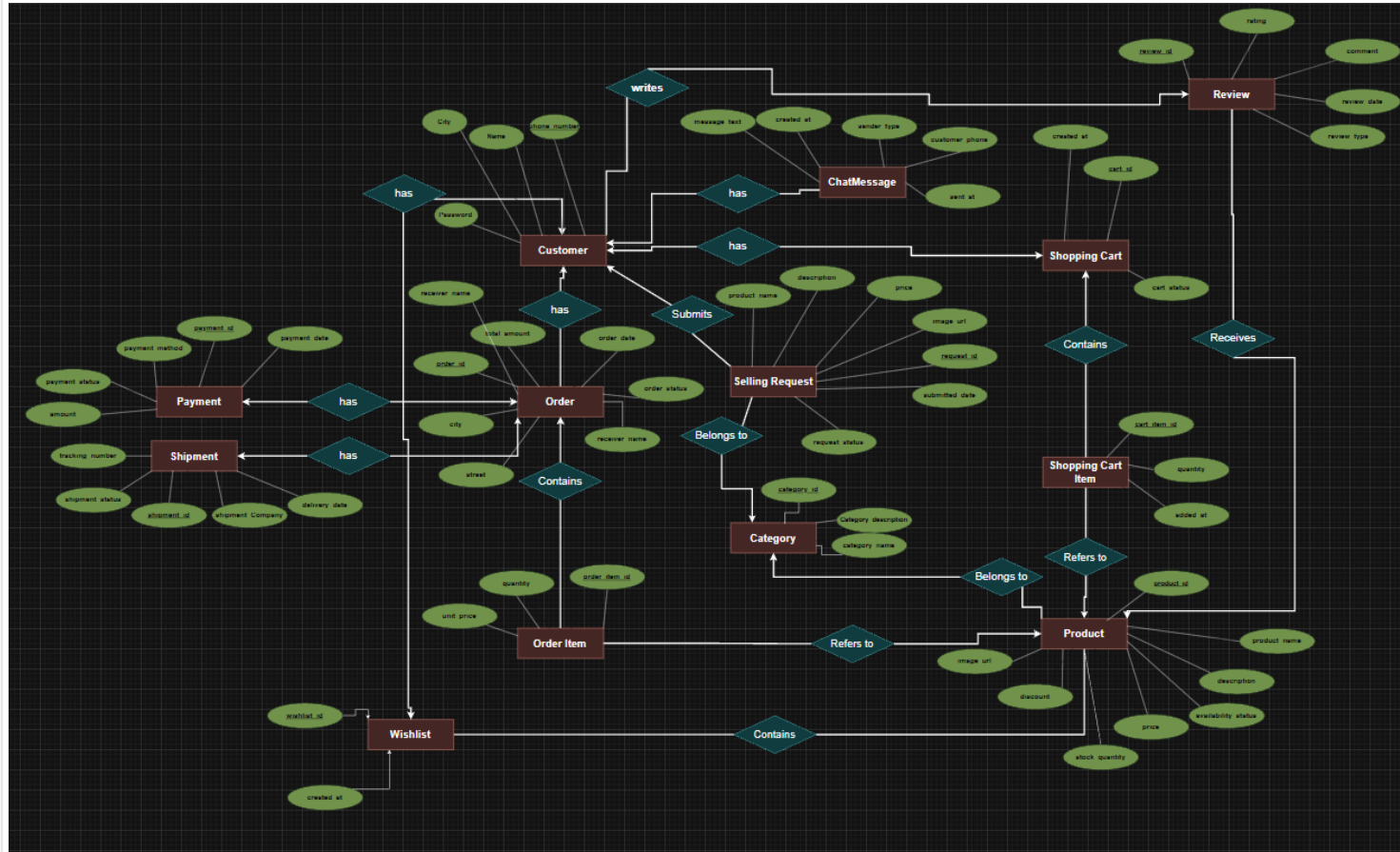
Statistics



Selling requests

# Database Design

## Main entities and relationships



ER diagram

## Core tables

Customer, Category, Product  
ShoppingCart + ShoppingCartItem  
Orders + OrderItem  
Payment + Shipment  
Review, Wishlist, SellingRequest,  
ChatMessage

2 central roles

13 main database tables

FK relationship driven design

# Normalization & Data Integrity

How repeated data and invalid values are reduced

## 1NF

Order products and cart products are not stored as repeated columns. They are separated into OrderItem and ShoppingCartItem.

## 2NF

Product data stays in Product instead of being repeated inside relationship tables.

## 3NF

Category, payment, shipment, and review details are separated into their own tables.

## Integrity rules used in SQL

Primary keys identify records.

Foreign keys connect related tables.

Unique constraints prevent duplicates.

CHECK constraints control statuses, ratings, prices, and stock.

Default values support dates and initial statuses.

```
CREATE TABLE onlinestore.Product (  
  product_id INT,  
  product_name VARCHAR(150) NOT NULL,  
  description VARCHAR(255),  
  price NUMERIC(10,2) NOT NULL,  
  image_url VARCHAR(255),  
  discount BOOLEAN DEFAULT 0,  
  availability_status VARCHAR(20) DEFAULT 'Available',  
  category_id INT NOT NULL,  
  stock_quantity INT NOT NULL DEFAULT 10,  
  created_at TIMESTAMP DEFAULT now(),  
  PRIMARY KEY (product_id),  
  FOREIGN KEY (category_id) REFERENCES onlinestore.Category(category_id),  
  CHECK (price >= 0),  
  CHECK (stock_quantity >= 0),  
  CHECK (availability_status IN ('Available', 'OutOfStock', 'Hidden'))  
);
```

SQL CHECK example

tings - Crucial DDR4 RAM 16GB

Average Rating: 4.5 / 5  
Total Reviews: 2

---

Customer: 0591234567  
Rating: 4 / 5  
Comment: It is good  
Date: 2026-05-16 17:59:50.0

---

Customer: 0599000001  
Rating: 5 / 5  
Comment: Very good RAM  
Date: 2026-05-16 14:43:42.0

---

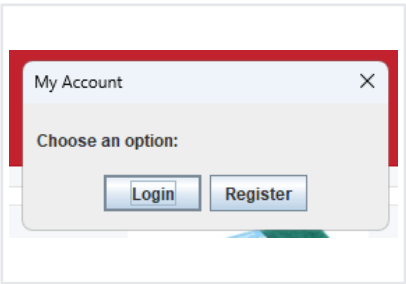
OK

Rating data

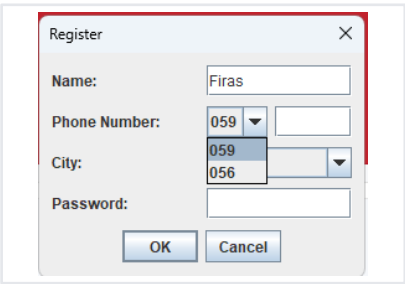
# Customer Workflow

From account creation to final order

- Register or log in with valid phone rules.
- Browse products by category or search/filter text.
- Add products to cart or wishlist.
- Checkout with receiver details and payment method.
- Confirm shipment and review products after delivery.
- Use support chat when help is needed.

A dialog box titled "My Account" with a close button (X) in the top right corner. It contains the text "Choose an option:" and two buttons: "Login" and "Register".

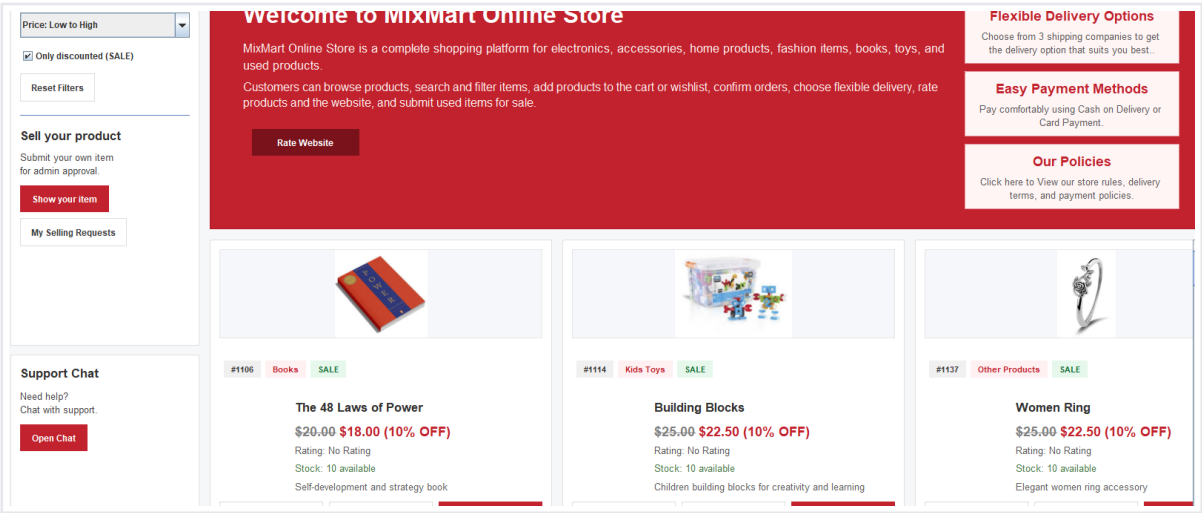
Account options

A "Register" dialog box with a close button (X) in the top right corner. It contains input fields for "Name:" (filled with "Firas"), "Phone Number:" (with a dropdown menu showing "059" and "056"), "City:" (with a dropdown menu showing "059" and "056"), and "Password:". There are "OK" and "Cancel" buttons at the bottom.

Register form

A "Filters" sidebar with a dropdown menu for "Price: Low to High". Below it is a list of filter options: "Default", "Highest Rating", "In Stock First", "Price: Low to High", and "Price: High to Low".

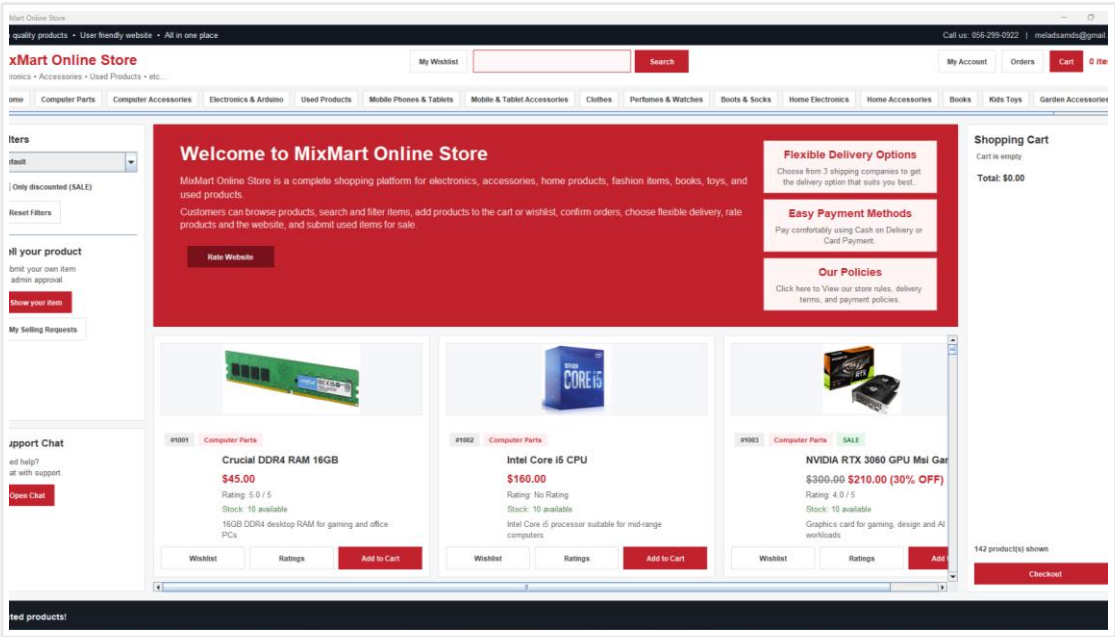
Sort/filter

A screenshot of the MixMart Online Store homepage. The header is red with the text "Welcome to MixMart Online Store". Below the header is a red banner with the text "MixMart Online Store is a complete shopping platform for electronics, accessories, home products, fashion items, books, toys, and used products." and "Customers can browse products, search and filter items, add products to the cart or wishlist, confirm orders, choose flexible delivery, rate products and the website, and submit used items for sale." There are three product cards displayed: "The 48 Laws of Power" (price \$20.00, \$18.00 (10% OFF)), "Building Blocks" (price \$25.00, \$22.50 (10% OFF)), and "Women Ring" (price \$25.00, \$22.50 (10% OFF)). The sidebar on the left contains a "Price: Low to High" dropdown, a "Only discounted (SALE)" checkbox, a "Reset Filters" button, a "Sell your product" section, and a "Support Chat" section.

Search result screen

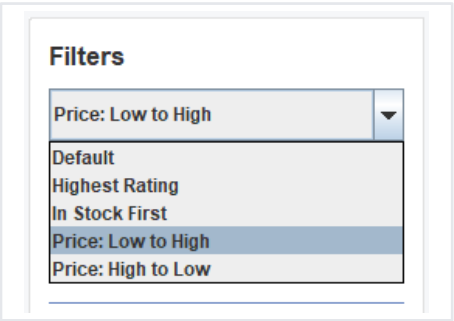
# Customer Side: Home, Search, Filters

Core browsing features used during the demo

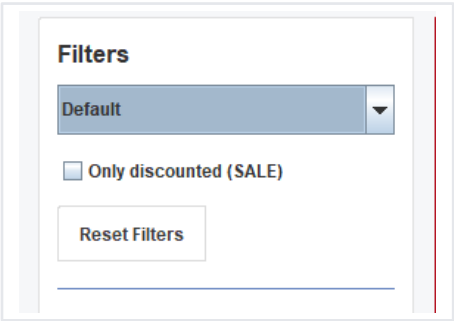


Home screen

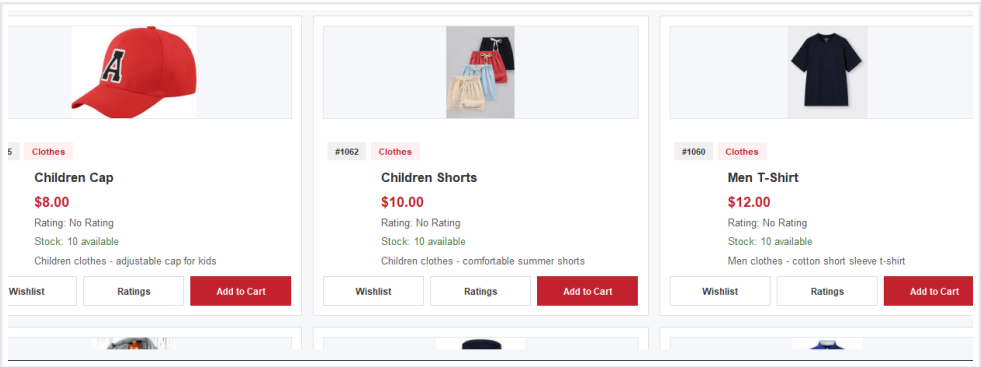
Demo focus: show category navigation, sorting, discount filter, reset filters, and search results.



Sorting



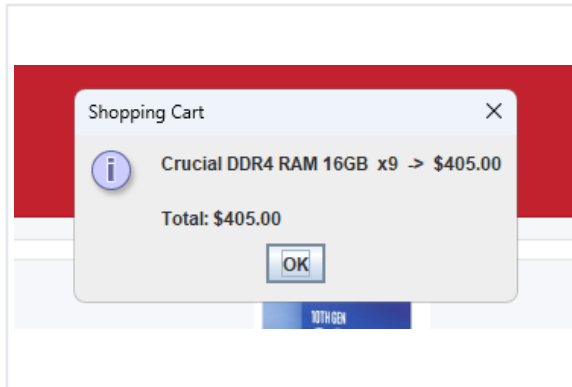
Discount filter



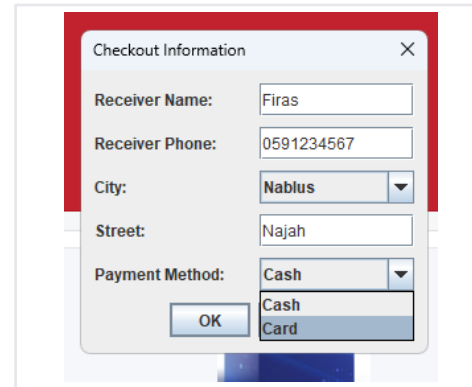
Category result

# Cart, Checkout & Orders

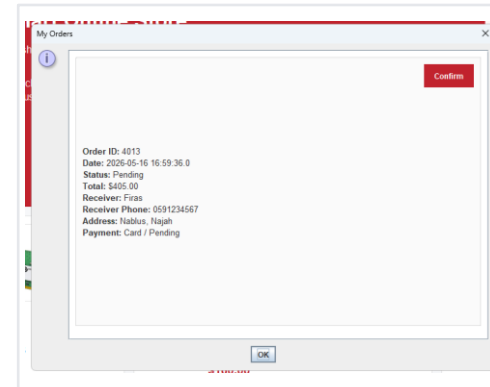
Database operations behind shopping actions



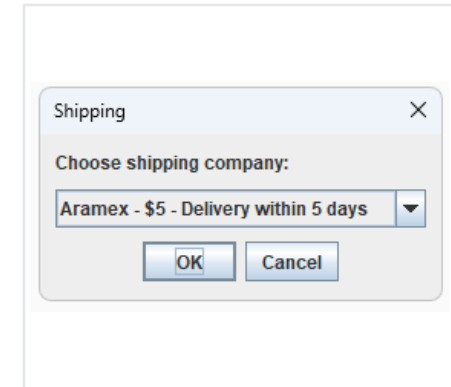
Cart summary



Checkout info



Order window



Shipping company

## What happens in the database?

Cart item quantity is updated.  
Checkout creates order and order items.  
Payment and shipment records are stored.  
Stock is reduced after successful order.  
Validation prevents buying beyond stock.

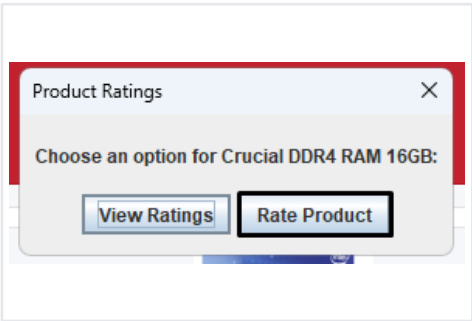
## Demo moment

Show adding a product, increasing quantity, checkout, and order confirmation.

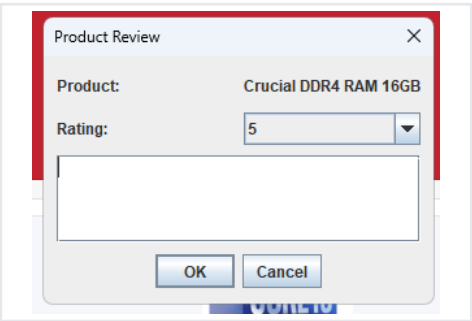


# Reviews & Selling Requests

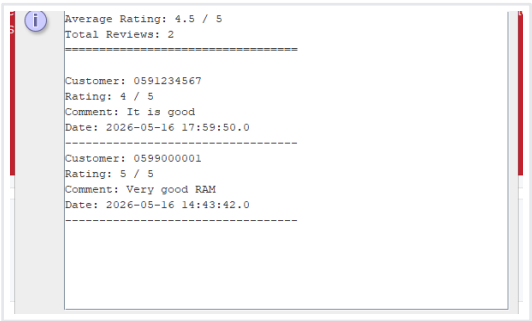
Features that connect customers with the store after purchase



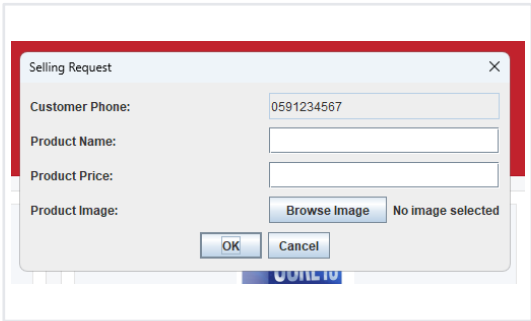
Review options



Product review form



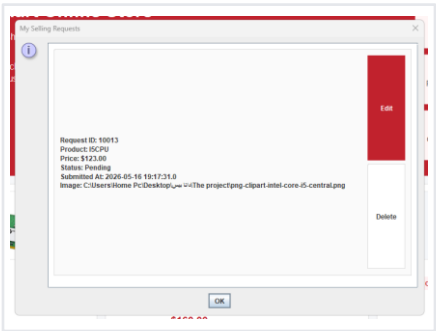
View ratings



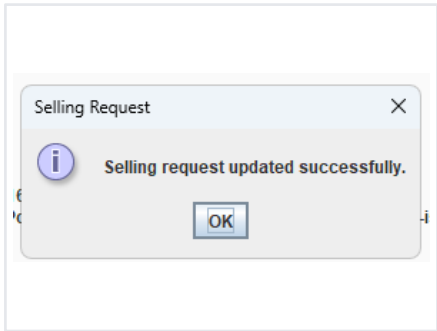
Selling request form

## Business rules

- Product reviews are allowed only after valid order status.
- Website reviews can be edited by the customer.
- Selling requests remain pending until admin approval.
- Pending requests can be edited or deleted by the customer.



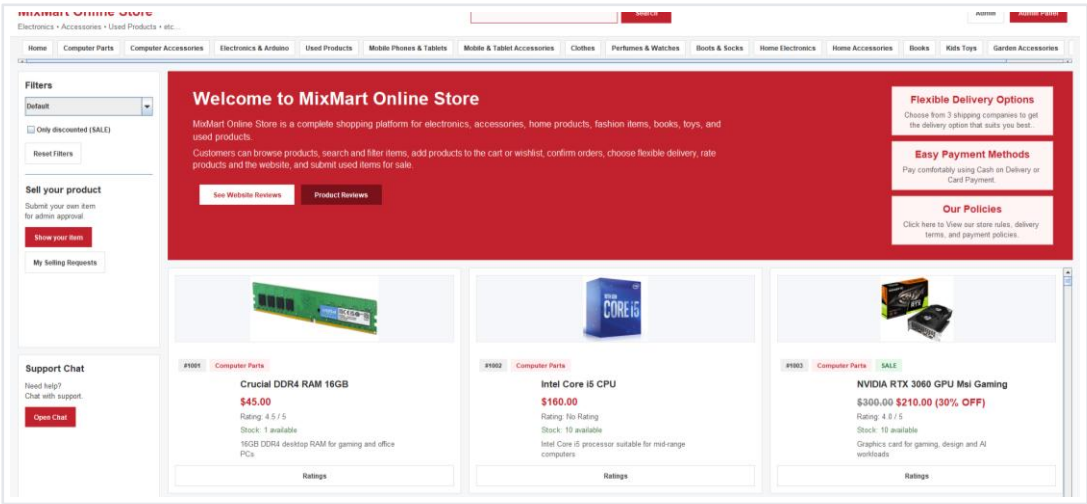
Submitted request



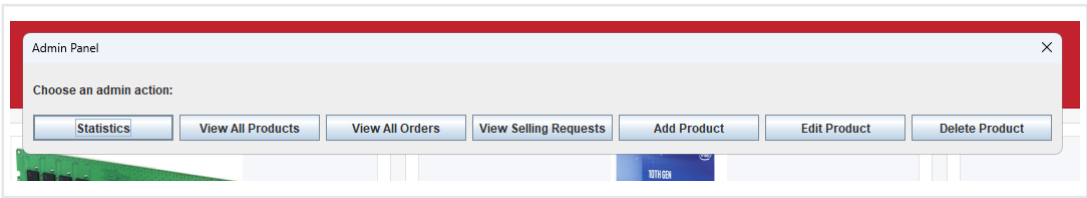
Request updated

# Admin Side: Monitoring & Control

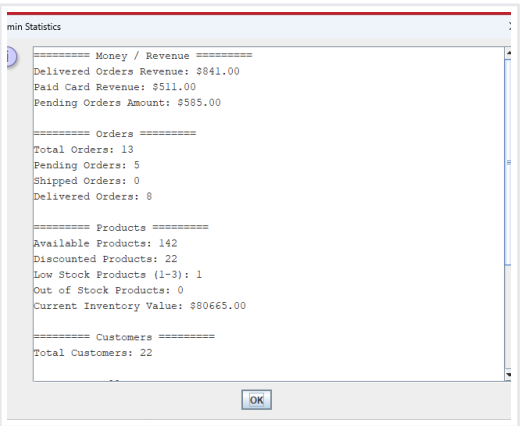
Admin features used to manage the store



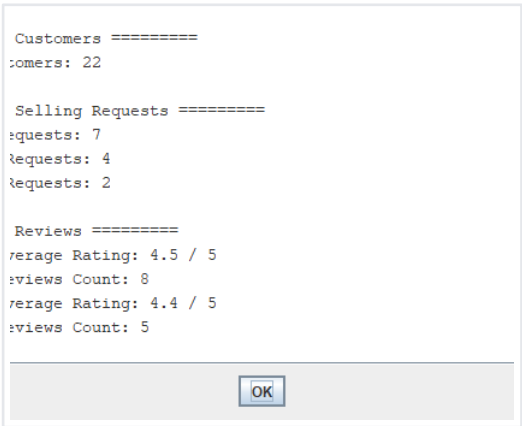
Admin home after login



Admin panel buttons



Statistics part 1

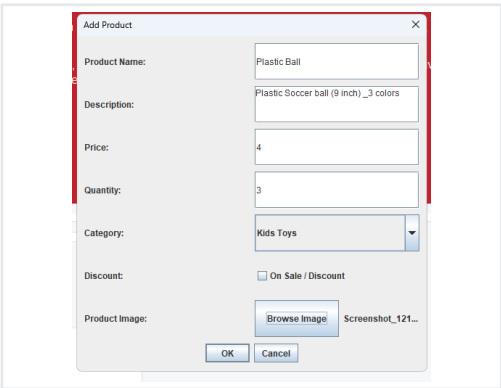


Statistics part 2

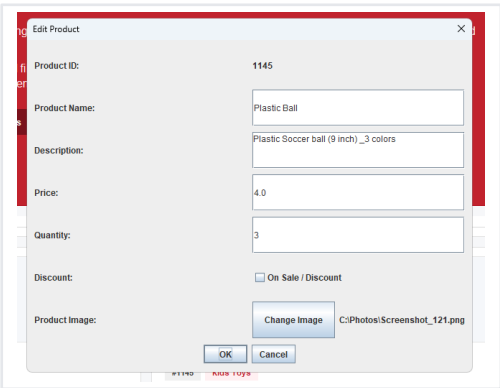
- View statistics
- Manage products
- View orders
- View reviews
- Accept/reject selling requests
- Support customers through chat

# Admin Product & Request Management

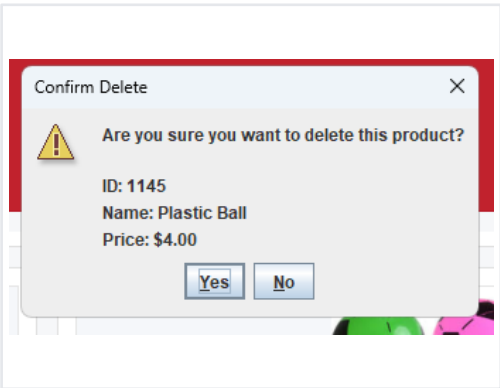
Adding, editing, hiding products and handling requests

A dialog box titled "Add Product" with fields for Product Name, Description, Price, Quantity, Category, Discount, and Product Image. The Product Name is "Plastic Ball", Description is "Plastic Soccer ball (9 inch)\_3 colors", Price is "4", Quantity is "3", Category is "Kids Toys", and Discount is "On Sale / Discount". The Product Image field has a "Browse Image" button and a file name "Screenshot\_121...".

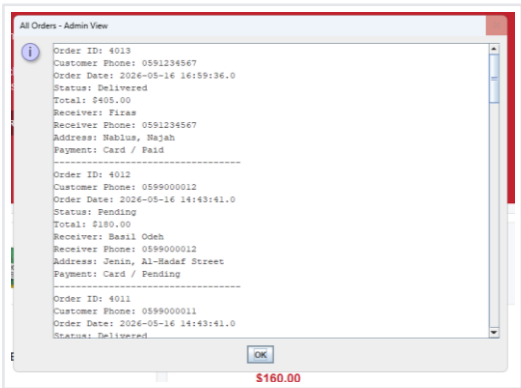
Add product

A dialog box titled "Edit Product" with fields for Product ID, Product Name, Description, Price, Quantity, Discount, and Product Image. The Product ID is "1145", Product Name is "Plastic Ball", Description is "Plastic Soccer ball (9 inch)\_3 colors", Price is "4.0", Quantity is "3", Discount is "On Sale / Discount", and Product Image has a "Change Image" button and a file name "C:\Photos\Screenshot\_121.png".

Edit product

A dialog box titled "Confirm Delete" with a warning icon and the text "Are you sure you want to delete this product?". It displays "ID: 1145", "Name: Plastic Ball", and "Price: \$4.00". There are "Yes" and "No" buttons.

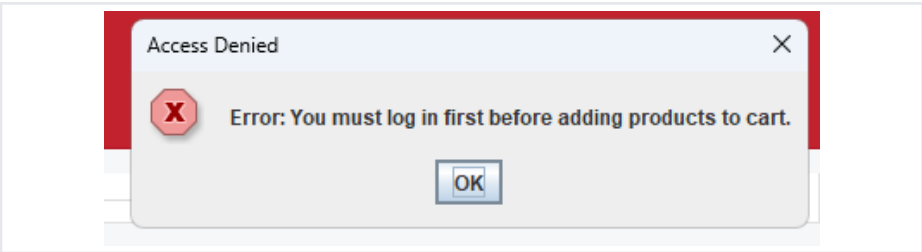
Confirm hide

A window titled "All Orders - Admin View" showing a list of orders. The first order is Order ID: 4013, Customer Phone: 0591234567, Order Date: 2024-05-16 14:59:34.0, Status: Delivered, Total: \$405.00, Receiver: Firas, Receiver Phone: 0591234567, Address: Habbus, MaJah, Payment: Card / Paid. The second order is Order ID: 4012, Customer Phone: 0599000012, Order Date: 2024-05-16 14:43:41.0, Status: Pending, Total: \$180.00, Receiver: Basil Odeh, Receiver Phone: 0599000012, Address: Jenin, Al-Radaf Street, Payment: Card / Pending. The third order is Order ID: 4011, Customer Phone: 0599000011, Order Date: 2024-05-16 14:43:41.0, Status: Delivered. The total amount shown at the bottom is \$160.00.

Selling requests

## Important implementation detail

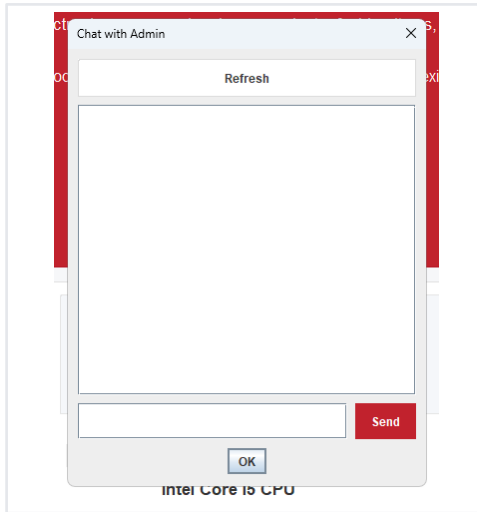
Deleting a product is implemented as hiding it using the availability status. This keeps historical order data safe and removes the item from customer views.

A dialog box titled "Access Denied" with a red "X" icon and the text "Error: You must log in first before adding products to cart." There is an "OK" button.

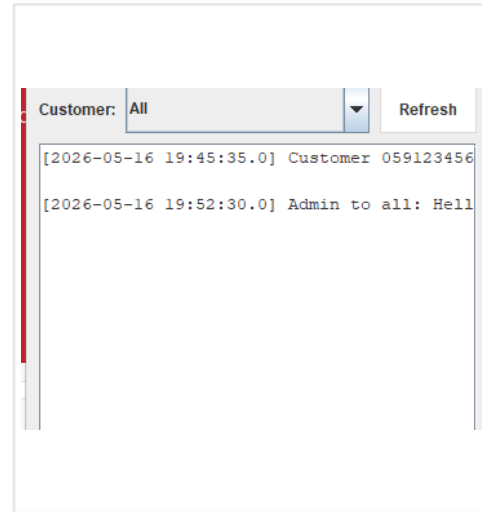
Product hidden message

# Support Chat

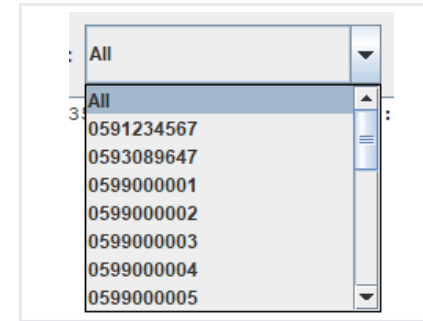
Customer-admin communication stored in the database



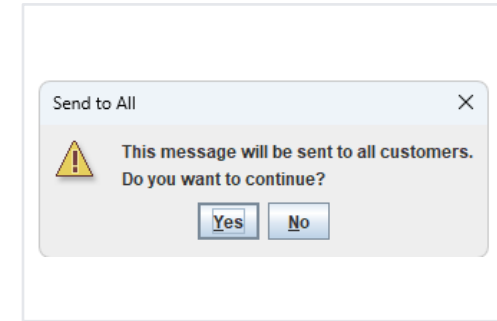
Customer chat



Admin chat



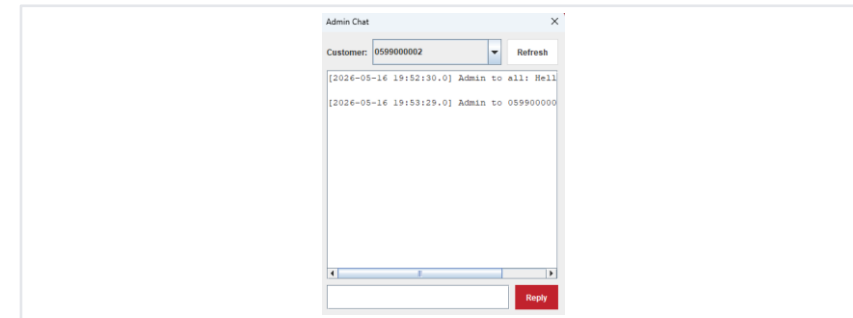
Select customer



Broadcast warning

## Chat logic

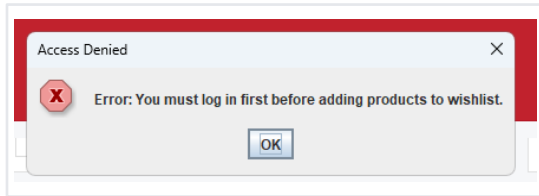
- Customers send messages to admin.
- Admin can reply to one customer.
- Admin can send one message to all customers.
- Messages are ordered by time and stored in ChatMessage.



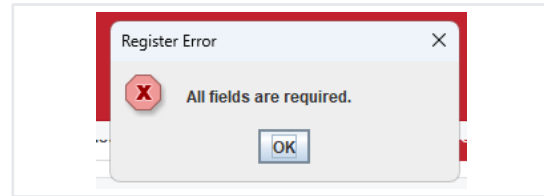
Admin reply screen

# Testing & Validation

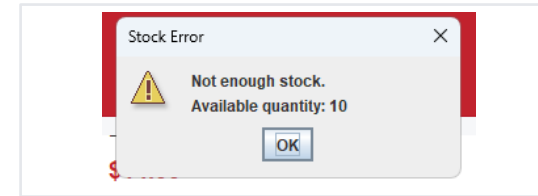
Manual tests for customer, admin, and database rules



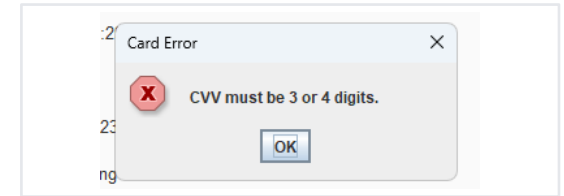
Login required



Required fields

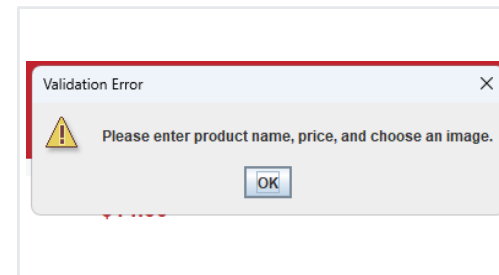


Stock error

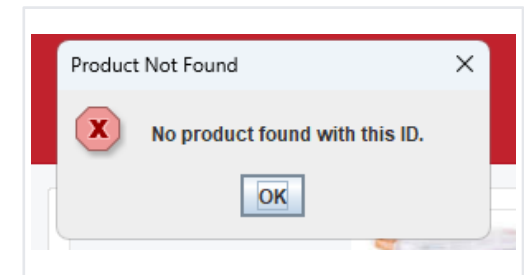


CVV error

Registration and login validation  
Cart quantity and stock validation  
Checkout receiver data validation  
Card information validation  
Admin add/edit/delete validation  
Selling request and chat testing



Add product error



Product not found

## Testing result

The system was checked through customer and admin scenarios to verify GUI behavior, SQL operations, relationships, and validation messages.

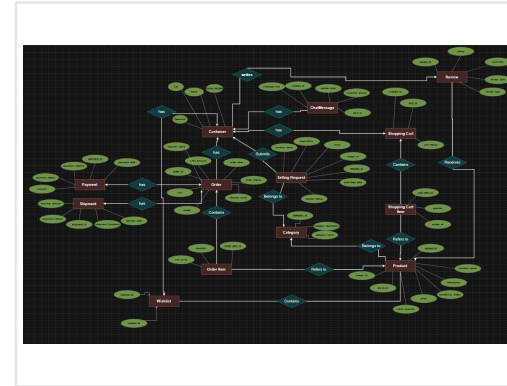
# Final Submission Package

Files that support the final delivery

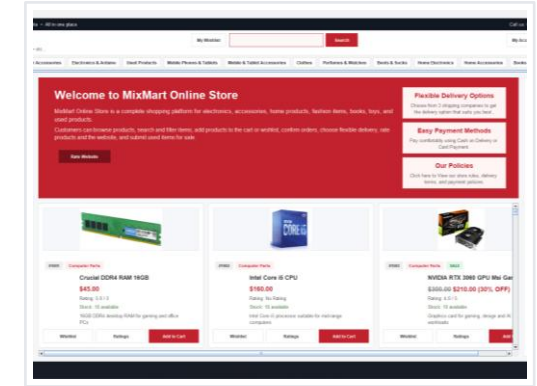
Java source code with GUI and DAO classes  
SQL script and database dump  
ER Diagram PDF/Image  
Normalization document  
Project report  
PowerPoint presentation  
Product images folder  
Recorded video demo with team participation

## Demo recording focus

The video should show the project from the idea to the final system, then demonstrate the customer side, admin side, database design, validation, and core features live in action.



ERD



GUI

```
CREATE TABLE onlinestore.Product (  
  product_id INT,  
  product_name VARCHAR(150) NOT NULL,  
  description VARCHAR(255),  
  price NUMERIC(10,2) NOT NULL,  
  image_url VARCHAR(255),  
  discount BOOLEAN DEFAULT 0,  
  availability_status VARCHAR(20) DEFAULT 'Available',  
  category_id INT NOT NULL,  
  stock_quantity INT NOT NULL DEFAULT 10,  
  created_at TIMESTAMP DEFAULT now(),  
  PRIMARY KEY (product_id),  
  FOREIGN KEY (category_id) REFERENCES onlinestore.Category(category_id),  
  CHECK (price >= 0),  
  CHECK (stock_quantity >= 0),  
  CHECK (availability_status IN ('Available', 'OutOfStock', 'Hidden'))  
);
```

SQL

```
===== Money / Revenue =====  
Delivered Orders Revenue: $841.00  
Paid Card Revenue: $511.00  
Pending Orders Amount: $585.00  
  
===== Orders =====  
Total Orders: 13  
Pending Orders: 5  
Shipped Orders: 0  
Delivered Orders: 8  
  
===== Products =====  
Available Products: 142  
Discounted Products: 22  
Low Stock Products (1-3): 1  
Out of Stock Products: 0  
Current Inventory Value: $90665.00  
  
===== Customers =====  
Total Customers: 22
```

Admin data

# Conclusion

What the project demonstrates

## Mix Mart Online connects database theory with a working Java application.

ER modeling and relational schema design

Normalization and SQL constraints

JDBC and DAO-based database connectivity

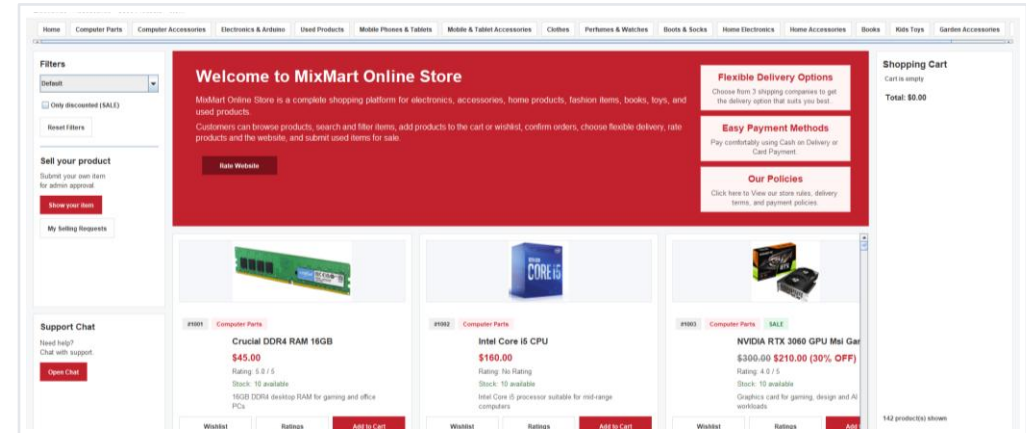
Customer and admin workflows

Testing, validation, and final demo preparation

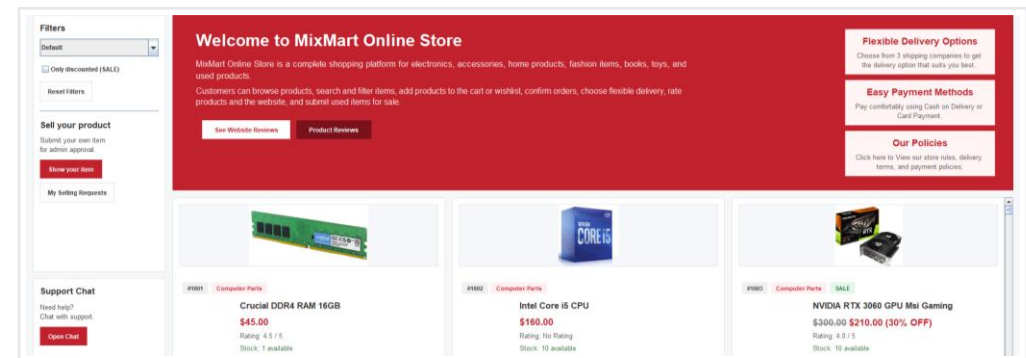
# Thank you

Melad Sam | Emad Ataout

Mix Mart Online | Final Project Demo



Final customer GUI



Final admin GUI